

		$= 3.0 \times 10^{-14} \text{ J}$
30	D	After 3.0×10^{12} nuclei decayed, the number remaining is 2.0×10^{12} $N = N_0 e^{-\lambda t}$ $2.0 \times 10^{12} = 5.0 \times 10^{12} e^{-(1.15 \times 10^{-8})t}$ $t = 7.97 \times 10^7 \text{ s}$